START HERE · 00–05 MINUTES

# Key ideas before the questions

Pure Mathematics + Linear Algebra · Week 5 · Keep this page beside your working.

## 1 · Functions: always state the domain and codomain

For $f : X \to Y$: **injective** means $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$; **surjective** means every $y \in Y$ has some $x \in X$ with $f(x) = y$. **Bijective** means both.

For $A \subseteq X$, $B \subseteq Y$:

$$f[A] = \{f(x) : x \in A\}, \qquad f^{-1}[B] = \{x \in X : f(x) \in B\}.$$

An image follows inputs forward; a preimage collects *all* inputs landing in the target set. Preimages exist even when no inverse function exists.

For $f : X \to Y$, $g : Y \to Z$, $(g \circ f)(x) = g(f(x))$: apply $f$ first. The composition of two surjections is surjective.

**Proof checklist:** injectivity → start with equal outputs; surjectivity → choose an arbitrary target and construct a legal input; set equality → prove both inclusions; a false universal claim → give one valid counterexample.

**Watch:** $f[A \cap B] \subseteq f[A] \cap f[B]$ always; equality holds if $f$ is injective. Changing the codomain can change surjectivity.

## 2 · Vector spaces: always state the scalar field

For $v_1, \ldots, v_k \in V$ over a field $K$:

$$\operatorname{span}_K\{v_1, \ldots, v_k\} = \left\{ \sum_{j=1}^{k} a_j v_j : a_j \in K \right\}.$$

**Independent:** $\sum a_j v_j = 0$ forces every $a_j = 0$. **Dependent:** some such relation has at least one nonzero coefficient. A **basis** is both independent and spanning; $\dim_K V$ is the number of vectors in any basis.

If $\dim V = n$: fewer than $n$ vectors cannot span; more than $n$ are dependent; exactly $n$ form a basis if either independence or spanning is proved. Size alone is not enough.

**Find a basis:** parameters → separate coefficients and check independence; equations → solve with free parameters; a spanning set → remove redundant vectors, or take pivot columns from the *original* matrix. To extend an independent set, add vectors outside its current span.

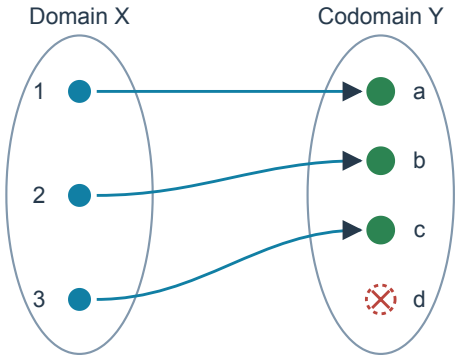
**Watch:** ambient coordinate count is not subspace dimension. Real and complex scalars give different independence tests. The zero space has empty basis, not $\{0\}$.

VISUAL REVIEW 1 · FUNCTIONS · USE WITHIN THE OPENING 5 MINUTES

# Follow the arrows

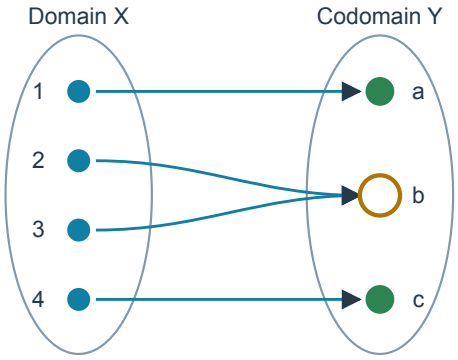
Every input must have **exactly one** arrow. Then ask: do inputs collide, and is any target missed?

## Injective only



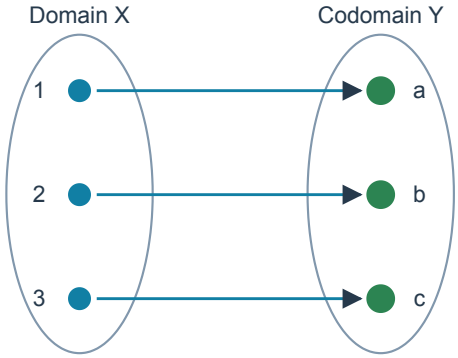

No collision, but one target is missed.
**Injective: yes · Surjective: no**

## Surjective only



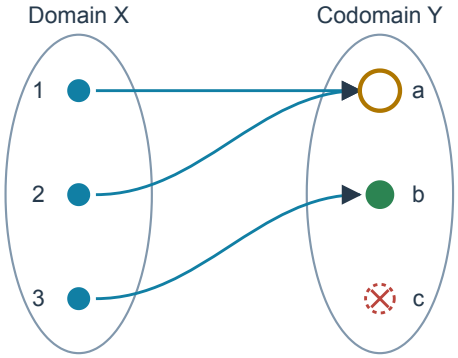

Every target is hit, but two inputs collide.
**Injective: no · Surjective: yes**

## Bijective




Every target has exactly one preimage.
**Injective: yes · Surjective: yes**

## Neither




A collision and a missed target.
**Injective: no · Surjective: no**

Blue arrows = function values · Yellow ring = collision · Dashed ring with × = missed target.

**Injective:** each target has at most one preimage.
**Surjective:** each target has at least one preimage.
**Bijective:** each target has exactly one preimage.

**Ask the room: “If we remove the unused target from the first codomain, what changes?”**

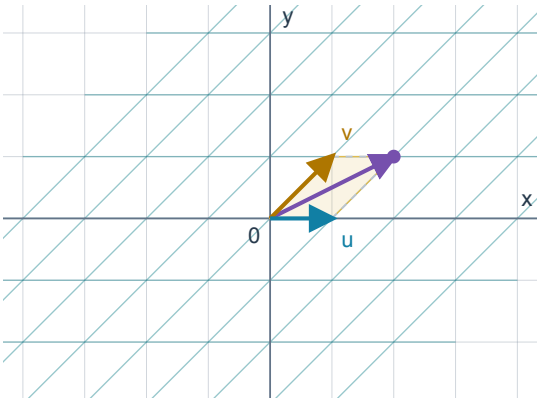
These are separate finite examples. All displayed targets belong to the stated codomain; changing the codomain changes the question.

VISUAL REVIEW 2 · LINEAR ALGEBRA · ALL SCALARS ARE REAL

# Two vectors: plane or line?

Fix $u = (1, 0)$, let $v = (1, t)$, and form $au + bv = (a + b, bt)$.

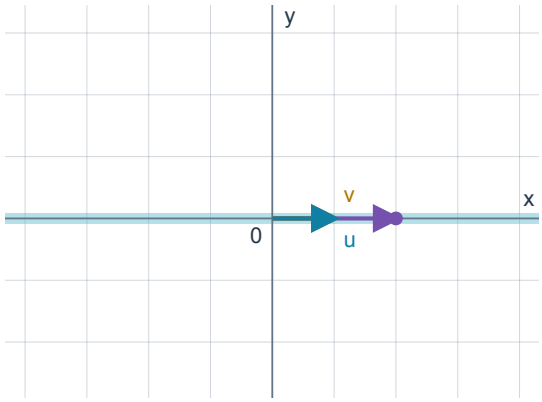
### Independent: t = 1




Span: the whole plane

The two directions generate every point of $\mathbb{R}^2$. Each point has unique coefficients.

### Collinear: t = 0




Span: one line through 0

Now $u = v$. Only the horizontal axis is reached, and coefficients are not unique.

Blue = u · Yellow = v · Purple = a u + b v · Dashed lines = vector addition. The highlighted parallelogram collapses when t = 0. The grid is only a visible sample; the span is unbounded.

### The bridge between both courses

Consider the coefficient map $T : \mathbb{R}^2 \to \mathbb{R}^2$, $T(a, b) = au + bv$.

Independent vectors ⇔ **injective** T: unique coefficients.
Spanning $\mathbb{R}^2$ ⇔ **surjective** T: every target is reached.
A basis of $\mathbb{R}^2$ ⇔ **bijective** T: both properties.

**Vector space check.** A subspace of $\mathbb{R}^2$ contains 0 and is closed under addition and scalar multiplication. A line through 0 is a subspace; the line $y = 1$ is not, since it omits 0.

**At t = 0:** the list $(u, v)$ is dependent. The singleton $\{u\}$ is a basis of its one-dimensional span, but it is not a basis of $\mathbb{R}^2$.

**Ask the room: “At t = 0, compare T(1,0) and T(0,1). Can any input reach (0,1)?”**

# Pure Mathematics

**MAIN ROUTE · 05–08 elapsed · 3 min**

## P1 · Quick diagnosis: injective versus surjective

Pure PS5 A9(a–c), A10(a–c)

**A9.** Let $f : \mathbb{R} \to \mathbb{R}^2$, $f(x) = (2x, x + 3)$. Find $f(0), f(1)$; prove that $f$ is injective; show that $(4, 4)$ is not attained.

**A10.** Let $h : \mathbb{R}^2 \to \mathbb{R}$, $h(x, y) = x + y$. Find $h(0, 1), h(2, 3)$; use $(0, 0), (-1, 1)$ to test injectivity; prove surjectivity.

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**MAIN ROUTE · 08–12 elapsed · 4 min**

## P2 · A proof using coordinates

Pure PS5 B3(a–c)

Let $F : \mathbb{R}^2 \to \mathbb{R}^3$ be $F(x, y) = (x + y, x - y, x^2 + y^2)$. Prove that $F$ is injective and not surjective. Is it bijective?

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**MAIN ROUTE · 12–17 elapsed · 5 min**

## P3 · Images and preimages: horizontal projection

Pure PS5 A2(a–b)

Let $f : \mathbb{Z}^2 \to \mathbb{Z}^2$, $f(m, n) = (m, 0)$, and

$$A = \mathbb{Z}_{>0} \times \{0\}, \quad B = \{0\} \times \mathbb{Z}, \quad C = \{(n, n) : n \in \mathbb{Z}\}.$$

Find $f[A], f[B], f[C]$, and $f^{-1}[A], f^{-1}[B], f^{-1}[C]$.

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**MAIN ROUTE · 17–23 elapsed · 6 min**

## P4 · Why images do not always preserve intersections

Pure PS5 A5(a–c), B6, B7

**A5.** For $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$, let $A = (-1, 0)$, $B = (0, 1)$. Find $f[A \cap B]$, $f[A]$, $f[B]$, and prove strict containment of the first set in $f[A] \cap f[B]$.

**B6.** If $f : X \to Y$ is not injective, construct $A, B \subseteq X$ with $f[A \cap B] \subsetneq f[A] \cap f[B]$.

**B7.** If $f$ is injective, prove equality for all $A, B \subseteq X$.

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**MAIN ROUTE · 23–26 elapsed · 3 min**

## P5 · Surjective maps compose

Pure PS5 A11

Suppose $f : X \to Y$ and $g : Y \to Z$ are surjective. Prove that $g \circ f : X \to Z$ is surjective.

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**MAIN ROUTE · 26–29 elapsed · 3 min**

## P6 · Four concrete image–preimage examples

Pure PS5 A6(a–d)

For $f : \mathbb{Z} \to \mathbb{Z}$, $f(x) = x^2$, find sets satisfying:

(a) $U_1 \subsetneq f^{-1}[f[U_1]]$; (b) $f[f^{-1}[V_1]] \subsetneq V_1$; (c) $U_2 = f^{-1}[f[U_2]]$; (d) $f[f^{-1}[V_2]] = V_2$.

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# Linear Algebra

**MAIN ROUTE · 29–31 elapsed · 2 min**

## LA1 · Independence, spanning and a basis

Linear Algebra tutorial guide, Week 5, pp. 12–13; examples supplied for the guide's discussion prompts.

What do linear independence, spanning and a basis mean? Give an example in each of the four boxes below, with ambient space $\mathbb{R}^2$: independent/spanning, independent/not spanning, dependent/spanning, dependent/not spanning. Why are both conditions needed for a basis?

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**MAIN ROUTE · 31–34 elapsed · 3 min**

## LA2 · Three dependence tests — and what counts as proof

Linear Algebra tutorial guide, Week 5, p. 12, examples (i)–(iv).

Decide whether each family is linearly independent over $\mathbb{R}$. Give an explicit relation if it is dependent.

$$\text{(i)}\ \begin{pmatrix}1\\0\end{pmatrix},\ 0; \qquad \text{(ii)}\ \begin{pmatrix}1\\0\end{pmatrix},\ \begin{pmatrix}3\\0\end{pmatrix}; \qquad \text{(iii)}\ \begin{pmatrix}1\\2\end{pmatrix},\ \begin{pmatrix}1\\0\end{pmatrix},\ \begin{pmatrix}0\\2\end{pmatrix}.$$

Invent one independent family and one dependent family. When can an example settle a mathematical claim?

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**MAIN ROUTE · 34–37 elapsed · 3 min**

## LA3 · Find a basis from parameters

Linear Algebra tutorial guide, Week 5, p. 12, first basis example; real parameter domain made explicit.

Find a basis and the dimension of

$$V_1 = \{(x, y, 2x, x+y)^T : x, y \in \mathbb{R}\} \subseteq \mathbb{R}^4.$$

Give a different basis as well.

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**MAIN ROUTE · 37–39 elapsed · 2 min**

## LA4 · Find a basis from an equation

Linear Algebra tutorial guide, Week 5, p. 12, second basis example.

Find two bases and the dimension of

$$V_2 = \{(x, y, z)^T \in \mathbb{R}^3 : z = x + 2y\}.$$

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**MAIN ROUTE · 39–43 elapsed · 4 min**

## LA5 · Find a basis from a spanning set

Linear Algebra tutorial guide, Week 5, p. 12, third basis example; renamed V3 because the source repeats V2.

Find a basis and the dimension of

$$V_3 = \operatorname{span}_{\mathbb{R}} \left\{ \underbrace{\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}}_{a}, \underbrace{\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}}_{b}, \underbrace{\begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}}_{c} \right\}.$$

Can more than one subset of this spanning set be a basis?

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**MAIN ROUTE · 43–46 elapsed · 3 min**

## LA6 · Three bases over the complex numbers

Linear Algebra tutorial guide, Week 5, p. 12, warm-up.

Write down three different bases for the complex vector space

$$V = \{(x, y, x + y)^T : x, y \in \mathbb{C}\} \subseteq \mathbb{C}^3.$$

State its dimension over $\mathbb{C}$.

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**MAIN ROUTE · 46–49 elapsed · 3 min**

## LA7 · What size tells us in R8

Linear Algebra tutorial guide, Week 5, p. 13; the source's condition |S|<7 is preserved, then strengthened explicitly.

Let $S \subseteq \mathbb{R}^8$. What can we say if (a) $|S| < 7$, (b) $|S| \geq 9$, or (c) $|S| = 8$? What changes at $|S| = 7$?

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MAIN ROUTE · 49–52 elapsed · 3 min

## LA8 · Removing and adding vectors carefully

Linear Algebra tutorial guide, Week 5, p. 13, basis reduction/extension discussion; concrete examples supplied.

(a) From the spanning set $S = \{e_1, e_2, e_1 + e_2\}$ for $\mathbb{R}^2$, remove a vector to obtain a basis. Why is arbitrary removal unsafe in general?

(b) Extend the independent set $T = \{(1, 1, 0)^T\}$ to a basis of $\mathbb{R}^3$. What property must each newly added vector have?

# After-class practice

Choose one reserve problem if you finish early.

RESERVE / LOOKUP

## P7 · Cartesian products and strict containment

Pure PS5 A1

Find integer subsets $X_1, X_2, Y_1, Y_2$ such that

$$(X_1 \times Y_1) \cup (X_2 \times Y_2) \subsetneq (X_1 \cup X_2) \times (Y_1 \cup Y_2).$$

RESERVE / LOOKUP

## LA9 · Reserve: intersecting bases is not intersecting spans

Linear Algebra tutorial guide, Week 5, p. 13, challenging question and footnote 7; missing y parameter in source footnote corrected.

Let $V, W$ be subspaces of $\mathbb{R}^3$, with respective bases $B_V, B_W$. Must $B_V \cap B_W$ be a basis of $V \cap W$? Prove it or give a counterexample.

Adapted from Pure Mathematics Problem Sheet 5 and the Week 5 Linear Algebra tutorial guide. Teacher solutions and pending hand-in solutions are excluded.